

Sambit K. Giri

Department of Astronomy – Stockholm University
Roslagstullsbacken 21, 106 91 Stockholm, Sweden

☎ +46 790243943 • ✉ sambit.giri@gmail.com, sambit.giri@astro.su.se
🌐 sambit-giri.github.io/ • orcid.org/0000-0002-2560-536X

Research Interests

I am an astrophysicist developing advanced **data analysis methods** — integrating **physics-informed AI**, **differentiable modeling**, and **topological data analysis** — to extract fundamental physics from next-generation astronomical surveys. My work bridges high-dimensional theoretical simulations and complex observational data from facilities such as **Low-frequency Array (LOFAR)**, the **Square Kilometre Array (SKA)**, **Euclid**, and the **Nancy Grace Roman Space Telescope**. I build scalable, physics-integrated emulators and inference frameworks that enable rigorous interpretation of these datasets, with a focus on **cosmic reionization** and **dark matter**. My vision is to establish a research program that treats **machine learning** as a rigorous mathematical extension of theoretical modeling, aiming to uncover the fundamental physical laws governing the **Cosmic Dawn**.

Academic Employments

After my PhD defence, I have been employed as a postdoctoral researcher at the following research institutes.

- **Department of Astronomy, Stockholm University** **Stockholm, Sweden**
Senior Researcher *Sep 2025–Present*
- **Van Swinderen Institute, University of Groningen** **Groningen, Netherlands**
Postdoctoral fellow *Oct 2024–Aug 2025*
- **Nordic Institute for Theoretical Physics (NORDITA)** **Stockholm, Sweden**
Nordita fellow *Oct 2022–Sep 2024*
- **Department of Astrophysics, University of Zurich** **Zurich, Switzerland**
Postdoctoral researcher *Jan 2020–Sep 2022*
- **Department of Astronomy, Stockholm University** **Stockholm, Sweden**
Postdoctoral researcher *Apr 2019–Dec 2019*

Education

Below I list education in decreasing chronological order.

- **Department of Astronomy, Stockholm University** **Stockholm, Sweden**
Doctorate in astronomy, Supervisor: Prof. Garrelt Mellema *Oct 2015 – Apr 2019*
Thesis: Tomographic study of the 21 cm signal during reionization
- **Indian Institute of Technology (Banaras Hindu University)** **Varanasi, India**
Integrated master of technology, Engineering physics *Jul 2010 – Oct 2015*
Gold Medallist, equivalent to summa cum laude
Master's thesis: Study of dynamic events on the solar photosphere
Bachelor's thesis: Diagnostics of Magnetohydrodynamic (MHD) Waves

Achievements

Scientific.....

- o In total, I have authored **82 publications**, including 73 in international peer-reviewed journals, 4 astrophysics codes and 5 conference papers
- o I have made significant contribution to more than 80% of these publications, including **16 as first author, 17 as second author, 8 as third author**
- o Among the second and third author contributions, **15 were led by students and junior researchers** whom I had the privilege to mentor
- o Citation metrics. ***h*-index: 31, *m*-index: 3.5, *i*10-index: 55, *i*100-index: 5** (retrieved from Google Scholar on 26 June 2026). The most recent citation data and its evolution over the years can be found at the following links: [SAO/NASA Astrophysics Data System](#) and [Google Scholar](#).
- o Completed my PhD 6 months before the scheduled time
- o I have been a **regular reviewer** for the following journals, which shows that I have extensive knowledge of the field to be able to give useful feedback on novel works.
 - Since 2020: Journal of Open Source Software, Monthly Notices of the Royal Astronomical Society
 - Since 2022: Journal of Cosmology and Astroparticle Physics
 - Since 2023: Physical Review Letters
 - Since 2024: Astronomy & Astrophysics, Nature Communications Physics, Physical Review D
 - Since 2025: The Astrophysical Journal
- o I have served as a **reviewer for research infrastructure proposals**, evaluating the scientific merit and computational requirements of proposed projects
 - Since 2024: DiRAC High Performance Computing resource proposal (UK Research and Innovation).
 - Since 2025: GMRT Observing Time Allocation proposal (National Centre for Radio Astrophysics, India).

Awards and Fellowships.....

- o Awarded the **Young Scientist Award** by the International Union for Radio Science (URSI) in 2023
- o Awarded the **Nordita fellowship** for independent research since 2022
- o Awarded gold medal (equivalent to **summa cum laude**) during Integrated Master's degree (2015)
- o Summer Research fellowship from Indian Academy of Sciences (2013)
- o National Talent Search Examination, scholarship from the Government of India (2008)
- o Qualified for the Graduate Aptitude Test in Engineering (GATE), conducted by the Department of Higher Education (Government of India), to receive a scholarship during the master's degree programme

International Collaborations and Memberships

I am a part of the following large international collaborations in the field of astronomy.

- o **Square Kilometre Array (SKA)**: I play a key role in the 'Theory/Numerical Simulations', 'Signal Analysis and Interpretation' and 'Synergy' focus groups of the SKA CD/EoR science team. I build numerical simulations and predictions of the expected signal that are used to develop observational strategies for the low-frequency component of SKA. I develop tools to optimally extract information from the expected SKA image data, which are packaged into an open-source code called Tools21cm.
- o **Low-frequency Array (LOFAR)**: I am a core member of the LOFAR EoR science team. My simulations are used to develop the data analysis pipeline. I have developed the Bayesian inference framework to interpret the observations.

- **Euclid space mission:** I am involved in the theory and simulations working group. I built a framework for modelling the baryonic feedback process that will affect the Euclid weak lensing observations. I also develop dark matter models that can be constrained with these observations.
- **Nancy Grace Roman Space Telescope:** I am involved in building theoretical models and observation strategies for probing the era of the first stars (Cosmic Dawn) with the upcoming infrared survey telescope.

In addition to my collaborative work, I am an active member of several prestigious astronomical societies.

- **European Astronomical Society** (2024-Present)
- **International Astronomical Union** (2024-Present)

Open Source Software & Data Products

- **Tools21cm:** A Python package for analyzing and interpreting 21-cm signal data. It implements sophisticated data science techniques, including topological data analysis (Betti numbers) and feature identification for large-scale structures.
- **pyC²Ray:** A massive-parallel, GPU-accelerated radiative transfer framework for simulating Cosmic Reionization. Designed for high-performance computing (HPC) environments to enable scalable simulations.
- **BEoRN:** A fast simulation code designed to rapidly explore the high-dimensional parameter space of Cosmic Reionization. Optimised for data interpretation and parameter inference tasks.
- **BCemu** (Baryonic Correction emulator): A machine learning emulator developed to model baryonic feedback effects on the matter power spectrum. The latest version is built on JAX/Flax for differentiable modeling. Integrated into the official Euclid mission end-to-end pipeline to constrain dark matter models.
- Developed a machine learning framework to identify extreme Lyman continuum leaking early photon sources or galaxies at high redshifts
- Bayesian Inference Framework for LOFAR: Developed a Bayesian inference pipeline coupled with machine learning-based surrogate models (emulators). This framework was deployed to interpret the latest upper limits from the LOFAR Epoch of Reionization observations.
- **AstronomyCalc:** A Python toolkit designed for university-level teaching. It contains pedagogical notebooks for cosmology and data analysis calculations. Used as a primary teaching tool for my classroom instruction and for onboarding Bachelor's/Master's students.

Technical Skills

- **Programming:** Python, C, C++, Fortran, CUDA (GPU programming), MATLAB, IDL.
- **Data Science & ML:** JAX, Flax (Differentiable Programming), TensorFlow, PyTorch, Scikit-learn, Bayesian Inference (MCMC, Nested Sampling, SBI), CNNs, U-Nets, Topological Data Analysis.
- **HPC & Parallel Computing:** MPI, OpenMP, GPU acceleration, SLURM workload manager.
- **Development Tools:** Git/GitHub/GitLab (CI/CD), Docker, Jupyter ecosystem.

Grants and Computing Resources

I have successfully applied for several grants. This shows that different funding organisations find my work interesting. The computing times achieved were useful in carrying out useful simulations for current and upcoming astronomical exploration. With this experience, I will be able to gain ample computing time for my project, which help develop new models and data analysis methods. Below I list the approved grants and granted computing time in reverse chronological order.

Grants

- **Stockholm University Special Educational Funding (PI)** 2026
180k SEK (~17k EUR) for the project titled
Development of a Privacy-Preserving AI-based Assessment Assistant for Higher Education.
- **Olle Engkvists Stiftelse (PI)** 2024
1.8M SEK (~170k EUR) for the research project titled
Cosmic Reionization and Dark Matter: Unveiling the Connection.
- **Nordita's PhD Fellows visitor program (PI)** 2024
In-kind support provided to host a PhD student, Timothée Schaeffer
(University of Zurich), for a 1.5-month-long research visit
- **Wenner-Gren Foundations (co-I)** 2023
150k SEK (~15k EUR) for organising a 1-week-long conference,
namely [Cosmic Dawn at high latitudes Conference](#)
- **Royal Swedish Academy (co-I)** 2023
150k SEK (~15k EUR) for organising a 1-week-long conference,
namely [Cosmic Dawn at high latitudes Conference](#)
- **Nordita's PhD Fellows visitor program (PI)** 2023
In-kind support provided to host a PhD student, Anshuman Acharya
(Max Planck Institute for Astrophysics), for a 2-month-long research visit
- **Nordita workshop (co-PI)** 2023
600k SEK (~57k EUR) for organising a 4-week-long workshop on studying the era of
first stars and galaxies, namely [Cosmic Dawn at high latitudes](#)
- **Swedish Research Council (VR) Project grant (co-I)** 2021
3.12M SEK (~0.3M EUR) for 4-years to develop a data interpretation framework,
titled 'Explorations of the Epoch of Reionization'
- **GRC Travel Grants, University of Zurich (PI)** 2020
2.8k CHF (~2.8k EUR) for a two-month-long research visit
- **Knut and Alice Wallenberg's Foundation (PI)** 2018
10k SEK (~950 EUR) for a research visit
- **Alva and Lennart Dahlmark research grant (PI)** 2019
4.5k SEK (~430 EUR) for a research visit
- **Alva and Lennart Dahlmark research grant (PI)** 2018
10k SEK (~950 EUR) for attending a winter school
- **C F Liljevalch J:ors travel grant (as PI)** 2018
6k SEK (~570 EUR) for attending a conference
- **IAU travel grant (PI)** 2017
500 EUR for attending IAU symposium 333
- **Stockholm University donation stipend (PI)** 2018
12k SEK (~1.1k EUR) for attending a winter school
- **Stockholm University donation stipend (as PI)** 2017
12k SEK (~1.1k EUR) for attending a conference
- **Stockholm University donation stipend (PI)** 2016
11k SEK (~1.1k EUR) for attending a conference
- **Alva and Lennart Dahlmark research grant (PI)** 2016
14k SEK (~1.3k EUR) for a workstation

HPC & Computing Grants	
○ Swedish National Infrastructure for Computing (PI) 2,400 GPU-hours on Arrhenius for GPU-accelerated astrophysics simulations 480 GPU-hours on Dardel for GPU-accelerated astrophysics simulations	2026
○ Swedish National Infrastructure for Computing (co-PI) 24,000,000 core hours on Dardel for astrophysics projects at NORDITA 600,000 core hours on Tetralith for astrophysics projects at NORDITA 12,000,000 core hours on Dardel for running 3D RT cosmic reionization simulations 6,000(GPU)+100,000(CPU) hours on LUMI in Finland to upgrade our simulation framework	2024
○ Swedish National Infrastructure for Computing (co-PI) 24,000,000 core hours on Dardel for astrophysics projects at NORDITA 600,000 core hours on Tetralith for astrophysics projects at NORDITA 6,000,000 core hours on Dardel for running 3D RT cosmic reionization simulations	2023
○ Swiss National Supercomputing Centre (co-PI) 70,000 node hours for developing a deep learning algorithm for SKA image data	2022
○ Swedish National Infrastructure for Computing (Named collaborator) 450,000 core hours on Beskow for running 1D RT simulations	2018
○ Swedish National Infrastructure for Computing (Named collaborator) 3,600,000 core hours on Beskow for running 3D RT simulations	2016
○ Swedish National Infrastructure for Computing (Named collaborator) 4,800,000 core hours on Beskow for running 3D RT simulations	2015

Teaching and Supervision

Here I list my teaching experience, formal pedagogical training and supervision experience.

Teaching.....
To date, I have **taught 1 course at the university level** and acted as a teaching assistant for multiple courses. Below I list the supervision and teaching experiences in decreasing chronological order.

○ Cosmology <i>Primary teacher; 7.5 ECTS</i>	Stockholm University <i>Oct 2022–Jan 2023, Oct 2026–Jan 2027 (Scheduled)</i>
○ Advanced simulation methods in Natural Sciences <i>Teaching assistant; 5 ECTS</i>	University of Zurich <i>Feb 2022–Apr 2022</i>
○ Upgrading the laboratory materials of Astronomy Lab <i>Teaching assistant</i>	Stockholm University <i>Sep 2019–Dec 2019</i>
○ Mechanics Lab <i>Teaching assistant</i>	Indian Institute of Technology (BHU) <i>Jan 2015–Apr 2015</i>
○ Electromagnetic Theory & Wave Guides <i>Teaching assistant</i>	Indian Institute of Technology (BHU) <i>Aug 2014–Nov 2014</i>

Formal pedagogical training.....
To teach and supervise students effectively, I took the following pedagogical training, which helped me understand the learning framework and improve my teaching methods.

- **Pedagogy course on ‘Research Supervision - theory and practice’** (Stockholm University, 2023): I took this course to enhance my student supervision skills. It provided insights on effective supervision, including giving constructive feedback and improving communication. The course also addressed potential challenges faced by students, such as anxiety, harassment, and non-inclusion. I learned strategies and discovered available resources to assist students dealing with these issues.

- **Pedagogy course on ‘Professional development course: Teaching and Learning’** (Stockholm University, 2022): This course introduced theories explaining how we perceive and learn new concepts, along with methods for structuring university-level courses.
- **Workshop on ‘Design for Active Learning’** (Stockholm University, 2022): This workshop introduced several active learning methods, such as role-playing and the Jigsaw Classroom, which I found that students really appreciate.
- **Workshop on ‘Making the case for case-based teaching’** (Stockholm University, 2022): This workshop assisted me in implementing optimal practices in astronomy courses. I find this teaching method useful for helping students not only learn new complex concepts but also apply them effectively.
- **Workshop on ‘Use the right digital tool for your teaching’** (Stockholm University, 2022): In the post-COVID-19 pandemic era, online teaching has become an essential aspect of university education. Therefore, educators should acquire this valuable skill. This workshop introduced me to various tools that can optimise teaching and keep students engaged. These digital and active-learning strategies are directly transferable to data-science classrooms, where interactive coding environments and real-time feedback are crucial.

Supervision.....

I was involved in the supervision of numerous student projects (supervised **4 master’s thesis and 3 bachelor’s theses**, co-supervised **4 PhD theses, 6 master’s theses and 2 bachelor’s theses**). In multiple cases, the work led to journal publications. These experiences show that I will be able to supervise students.

○ Co-supervision of PhD students:

- Viktor Köhlin Lövfors (Stockholm University; Sep 2025–Ongoing)
- Olof Nebrin (Stockholm University; Jan 2023–Ongoing)
- Jozef Bucko (University of Zurich; Sep 2020–Mar 2024)
- Timothée Schaeffer (University of Zurich; Jan 2020–Aug 2024)

○ Supervision of visiting PhD students:

- Timo P. Schwandt (Sussex University; Sep 2023–Mar 2024)
- Anshuman Acharya (Max Planck Institute for Astrophysics; Apr 2023–Jul 2023)

○ Supervision of master’s students:

- Rikard Lesley (Stockholm University; Jan 2026–Jun 2026)
- Abdullah Sheriff (Stockholm University; Oct 2023–Dec 2024)
- Karin Kjellgren (Stockholm University; Jan 2023–Jun 2023)
- Chrishon Nilanthan (ETH Zurich; Oct 2020–May 2021)

○ Supervision of bachelor’s students:

- Suha Aham (Stockholm University; Apr 2026–Ongoing)
- Felix Vecchi (ETH Zurich; Mar 2022–Aug 2022)
- Christina Fakiola (ETH Zurich; Mar 2022–Aug 2022)

○ Supervision of visiting master’s students:

- Aniket Nath (National Institute of Science Education and Research, India; May 2024–Sep 2024)

○ Co-supervision of master’s students:

- Judith Deen (University of Groningen; Apr 2025–Ongoing)
- Michael Kovac (ETH Zurich; Nov 2021–Jun 2022)
- Fabian Hervás Peters (University of Zurich; Aug 2021–Jun 2022)

- Zhongnan Cai (University of Zurich; Aug 2020–Apr 2021)
- Jonathan Hubert (ETH Zurich; Jan 2020–Jun 2020)
- Ancel Larzul (École Normale Supérieure - Paris; Mar 2019–Aug 2019)
- o **Co-supervision of bachelor's students:**
 - Eric Fredriksson (Stockholm University; Aug 2018-Dec 2018)
 - Thomas Aldheimer (Stockholm University; Aug 2018-Nov 2017)

Organisation and Institutional Responsibilities

I have been actively engaged in various administrative responsibilities alongside my research endeavours. Below, I outline some of the key roles I have undertaken.

- o 4-5 Feb 2026: **Scientific and Local Organising Committee** member - *3rd National Sweden SKA Science Days*; I was involved in preparing the full program of scientific talks and organising the local bookings required for the conference.
- o Nov 2024–Jun 2025: **Dutch Theoretical Cosmology meeting organiser**; I am currently responsible for coordinating the organization of the national theoretical cosmology meetings, which are held in rotation in Amsterdam, Groningen, Leiden, and Utrecht.
- o 24-28 Jun 2024: **Scientific and Local Organising Committee** member - *Cosmic Dawn at high latitudes Conference*; I was involved in preparing the full program of scientific talks and organising the local bookings required for the conference.
- o 10 Jun-5 Jul 2024: **Organising Committee** member - 4-week long Nordita workshop on *Cosmic Dawn at high latitudes*; I was involved in finalising the participants for each week and the programme schedule.
- o Jan 2023–Sep 2024: **Nordita Astrophysics seminar organiser**, Nordic Institute for Theoretical Physics (Nordita), Sweden; I was responsible for the international speaker selection process, inviting and arranging their visit and organising the talk at the institute.
- o Oct 2022–Sep 2024: **Cosmology journal club organiser**, Department of Astronomy, Stockholm University, Sweden; I was responsible for organising weekly meetings dedicated to discuss recent publications.
- o 02 Feb 2023: **Local organising committee** member - '1st National SKA Science Day Sweden'; I was part of the team responsible for organising the meeting, which involved tasks such as coordinating lunch arrangements and providing technical support during the presentations.
- o 02–04 Feb 2022: **Primary conference organiser** - 'Learning the high-redshift Universe'; organised an online conference during the COVID-19 pandemic, providing early career scientists with a platform to share their work and foster meaningful interactions.
- o Jun 2021–Jun 2022: **Cosmology seminar organiser**, Institute for Computational Science, University of Zurich, Switzerland; I was responsible for the international speaker selection process, inviting and arranging their visit and organising the talk at the institute.
- o Sep 2020–Jun 2021: **Student seminar organiser**, Institute for Computational Science, University of Zurich, Switzerland; I was responsible for interacting with doctoral students at the institute and helping them prepare their presentations and speak in front of an audience.
- o Sep 2019–Dec 2019: **Upgrading the laboratory materials of Astronomy Lab**, Stockholm University, Sweden; I was responsible for developing a student-friendly software package that give them hands-on experience in calculating the astronomical processes.

Talks and Seminars

I have presented my work in numerous forums, which shows that I have the capability to properly communicate the findings of my research work. Here I list my invited (15+) and contributed (25+) talks.

Invited.....

- European Astronomical Society Annual Meeting**

◦ *Witnessing the Epoch of Reionization with the Square Kilometre Array*

Lausanne, Switzerland

29 Jun 2026
- Cosmology Seminar, ETH Zurich**

◦ *Robust Exploration of Patchy Reionization*

Zurich, Switzerland

07 Apr 2025
- Understanding the Epoch of Reionization II**

◦ *E pluribus unum - how much do galaxy details matter?*

Sesto, Italy

13-17 Jan 2025
- Dutch Theoretical Cosmology Meeting**

◦ *Towards resolving the S_8 tension*

Groningen, Netherlands

4 Oct 2024
- 2nd National Sweden SKA Science Day**

◦ *Exploring the epoch of reionization with SKA*

Gothenburg, Sweden

10-11 Sept 2024
- Indian Institute of Technology, Madras**

◦ *Probing Cosmological Models through the First Galaxies*

Chennai, India

26 Jul 2024
- Nordita Day of Open Doors**

◦ *Overview of research in the Astrophysics division of Nordita*

Stockholm, Sweden

20 Nov 2023
- Kapteyn Astronomical Institute, University of Groningen**

◦ *On the use of machine learning for understanding cosmic reionization*

Groningen, Netherlands

8 May 2023
- Understanding the epoch of cosmic Reionization**

◦ *Simulating the large-scale signal during reionization and cosmic dawn*

Sesto, Italy

6-10 March 2023
- Nordita astrophysics seminar**

◦ *Can baryonic feedback processes resolve the S_8 tension?*

Stockholm, Sweden

25 Jan 2023
- Laboratory of Astrophysics, EPFL**

◦ *Knowing cosmic reionization with images*

Lausanne, Switzerland

31 May 2022
- Kapteyn Astronomical Institute, University of Groningen**

◦ *Modelling baryonic effects in weak lensing observations*

virtual/online

28 Feb 2022
- Tata Institute of Fundamental Research, Mumbai**

◦ *Interpreting the high-redshift 21-cm signal observations*

virtual/online

14 Jan 2022
- Indian Institute of Technology, Hyderabad**

◦ *Modelling baryonic effects in cosmological surveys*

virtual/online

1 Sept 2021
- RISU (Reionizing-Indore-Stockholm-Uppsala) meeting**

◦ *Characterising the topology of ionized regions with the Betti numbers*

virtual/online

18 Jan 2021
- Institute for computational science, UZH**

◦ *Probing cosmic reionization with the 21-cm signal*

Zurich, Switzerland

27 Mar 2020
- Institute for Particle Physics and Astrophysics, ETH Zurich**

◦ *Tomographic study of 21-cm signal during reionization*

Zurich, Switzerland

10 Feb 2020
- Next-Generation Cosmology with Next-Generation Radio Telescopes: II**

◦ *Parameter inference from 21-cm images during reionization*

Sesto, Italy

27-31 Jan 2020
- Observing the First Billion Years of the Universe**

◦ *Image analysis techniques for 21-cm tomography of the CD-EoR*

Indore, India

20-24 Jan 2020

Contributed.....

- Cosmology in the Alps 2026**

○ *Probing Cosmic Reionization's Non-Gaussianity with 21-cm Fourier Phases*

Les Diablerets, Switzerland

16-20 Mar 2026
- 3rd Sweden SKA Science Day**

○ *Mapping the Growth of Ionized Bubbles: SKA's Window into Cosmic Reionization*

Stockholm, Sweden

4-5 Feb 2026
- AI for Science, KVA symposium**

○ *Learning the Early Universe: AI-Driven Discovery from the Future SKA Observatory*

Stockholm, Sweden

3-4 Sep 2025
- Cosmic Dawn at High Latitudes**

○ *Exploring Beyond Standard Cosmological Models during the Epoch of Reionization*

Stockholm, Sweden

24-28 Jun 2024
- LOFAR EoR KSP meeting**

○ *pyC²Ray: speeding up reionization simulations employing GPUs*

Groningen, Netherlands

24-26 Apr 2024
- Cosmology in the Alps 2024**

○ *Exploring Beyond Standard Cosmological Models during the EoR*

Les Diablerets, Switzerland

18-22 Mar 2024
- SU Astronomy Departmental Conference**

○ *Accelerating Radiative Transfer Simulations with GPUs*

Stockholm, Sweden

12 Dec 2023
- Euclid Cosmological Simulations SWG meeting 2023**

○ *Validation of baryonic feedback modelling methods*

Barcelona, Spain

15-17 Nov 2023
- First Nordic Cosmology Meeting**

○ *Probing Cosmological Models Through the Era of First Galaxies*

Stockholm, Sweden

23-25 Oct 2023
- OKC@15: Celebrating 15 years of Oskar Klein Centre**

○ *Is the JWST Hinting Towards a Cosmological Model Beyond the Standard?*

Stockholm, Sweden

17-19 Oct 2023
- XXXVth URSI General Assembly and Scientific Symposium**

○ *Imprints of early universe galaxy formation on the 21-cm signal at cosmic dawn*

Sapporo, Japan

19-26 Aug 2023
- Cosmology from Home**

○ *Detecting Beyond Standard Model Cosmology through EoR Observations*

virtual/online

3-14 Jul 2023
- Reionisation in the Summer**

○ *Constraining the nature of dark matter with high-redshift observations*

Heidelberg, Germany

26-30 Jun 2023
- LOFAR EoR KSP meeting**

○ *Symbolic learning of the large scale 21-cm bias*

Haifa, Israel (hybrid)

10 May 2023
- National SKA Science Day Sweden**

○ *Constraining cosmology with the 21-cm signal during reionization*

Stockholm, Sweden

02 Feb 2023
- SKACH Kick-off Meeting, ETH Zurich**

○ *Interpretation framework for SKA observations*

Zurich, Switzerland

24 May 2022
- Advances in cosmology through numerical simulations (MIAPbP)**

○ *Simulating cosmic reionization at very large-scales*

Garching, Germany

10 May 2022
- LOFAR EoR KSP meeting**

○ *Fast 21-cm signal modelling framework*

Paris, France

28 Apr 2022
- 4th Global 21-cm Workshop**

○ *Imprints of mixed dark matter in the 21-cm signal*

virtual/online

11 Oct 2021
- Swiss SKA Days, EPFL**

○ *Tools21cm: a user-friendly package to create mock SKA observations*

Lausanne, Switzerland

8 Sept 2021

- **SKA Science Meeting (Breakout session)**
Constructing emulators for reionization simulations
 - **PhD defence**
Tomographic study of the 21-cm signal during reionization
 - **Friday seminar (Department of astronomy)**
Tomographic study of the 21-cm signal during reionization
 - **Rise and shine: galaxies in the epoch of reionization**
Squeezed-limit bispectrum of 21-cm observations
 - **LOFAR EoR KSP meeting**
Astrophysical parameter estimation from 21-cm signal
 - **Licentiate defence**
Tomographic study of the 21-cm signal during reionization
 - **IAU Symposium 333**
Constraining the Lyman Continuum escape using machine learning
 - **LOFAR EoR KSP meeting**
Position dependent 21-cm power spectra during reionization
 - **SKA CD/EoR ST meeting**
Bubble identification in 21-cm tomography

Manchester, UK
12 Apr 2019

Stockholm, Sweden
03 Apr 2019

Stockholm, Sweden
08 Mar 2019

Strasbourg, France
18 Jun 2018

Groningen, Netherlands
01 Feb 2018

Stockholm, Sweden
20 Jan 2018

Dubrovnik, Croatia
02 Oct 2017

Haifa, Israel
04 Apr 2017

Pisa, Italy
14 Mar 2017

Experience of outreach activities

I am motivated to inspire the next generation of scientists to pursue a career in astronomy. In addition to engaging in informal conversations with high school students to address their curiosity about complex concepts, I have actively pursued formal opportunities, which I list below.

- **Nordita Day of Open Doors (2023)**: I represented the Astrophysics division of Nordita during this event, which hosted first-year university students from Stockholm and Uppsala county. I delivered an overview seminar followed by an informal discussion on the methods and objectives of astrophysical research. I utilised animations generated from my simulations to elucidate the intricate process of early galaxy formation during the discussion.
- I created a [new rendition of the 'Flammarion engraving'](#) with the help of Dr Hannah Ross. The Flammarion engraving is an allegory representing humanity's curiosity and fascination with the universe, as well as the desire to explore and understand it. Our version presented how the Square Kilometre Array (SKA) will aid in this pursuit of understanding the cosmos. This creation was featured in SKA observatory's magazine, [Contact issue 01](#).

References

- **Prof. Garrelt Mellema**, Department of Astronomy, Stockholm University;
Email: garrelt.mellema@astro.su.se; Phone: +46 8553 785 52
- **Dr. Aurel Schneider**, Department of Astrophysics, University of Zurich;
Email: aurel.schneider@uzh.ch; Phone: +41 78 735 5180
- **Dr. P. Daniel (Daan) Meerburg**, Van Swinderen Institute, Groningen University, Groningen;
Email: P.D.Meerburg@rug.nl; Phone: +31 50 363 4950

Publications

I have authored a total of **82 research outputs**, consisting of 73 peer-reviewed journal articles, 4 astrophysics codes, and 5 conference proceedings. I have played substantial roles in **16 articles as first author**, **17 as**

second author, and 8 as third author. Among the second and third author contributions, **15 were led by students and junior researchers** I mentored.

My academic impact is reflected in an *h-index* of **31** and *i10-index* of **55**, with a total of **3000 citations** (annual growth rate: *m-index* = 3.5). Citation data from Google Scholar on 26 June 2026. Current data: [NASA/ADS](#) and [Google Scholar](#).

Peer-Reviewed Journal Articles.....

1. Bucko, J., Nicola, A., Schneider, A., Kovač, M., **Giri, S. K.**, Reischke, R., Bulbul, E., Clerc, N., Liu, A., Bartalucci, I., Refregier, A., 2026. Baryonification IV: Constraining baryonic feedback with X-ray observations. Submitted to journal, [arXiv:2609.09144](#)
2. EoR/CD Science Working Group,...**Giri, S. K.**, ..., 2026. Overview of 21 cm Experiments at high redshift with SKA-Low. *Advancing Astrophysics with the SKA – II (AASKAII)*, Report number [AASKAII/Trott01](#)
3. Acharya, A., Bag, S., Berger, S., Bianco, M., Breitman, D., Crascall-Kennedy, L., Dasgupta, S., Datta, A., Douspis, M., Georgiev, I., **Giri, S. K.**, Gorce, A., Heneka, C., Hiegel, J., Ian Hothi, I., Kapahtia, A., Kern, N., Mahida, Y., Maity, B., Majumdar, S., Mériot, R., Mondal, R., Murray, S. G., Noble, L., Pal, S. K., Schneider, A., Semelin, B., Shaw, A. K., Shimabukuro, H., Thélie, E., Tripathi, A., Trott, C. M., 2026. Inferring cosmology and astrophysics from the high-redshift 21 cm signal with SKA-Low. *Advancing Astrophysics with the SKA – II (AASKAII)*, Report number [AASKAII/Acharya01](#)
4. Acharya, A., Bianco, M., Breitman, D., Chen, H., Datta, A., Diao, K., **Giri, S. K.**, Heneka, C., Kern, N., Liu, A., Mahida, Y., Majumdar, S., Pal, S. K., Ni, S., Pietschke, Y., Piras, D., Shaw, A. K., Shimabukuro, H., Sui, C., Tripathi, A., Zhao, X., 2026. Machine Learning and the SKA for Cosmic Dawn and the Epoch of Reionization. *Advancing Astrophysics with the SKA – II (AASKAII)*, Report number [AASKAII/Acharya02](#)
5. Bag, S., Bianco, M., Choudhuri, S., Dasgupta, S., Datta, A., Datta, K. K., Georgiev, I., **Giri, S. K.**, Hothi, I., Kapahtia, A., Majumdar, S., Mellema, G., Mishra, A., Pal, S. K., Feiyu Zhao, F., 2026. Imaging the 21-cm signal from the Cosmic Dawn & Epoch of Reionization and the connection with the Global Signal. *Advancing Astrophysics with the SKA – II (AASKAII)*, Report number [AASKAII/Bag01](#)
6. Barkana, R., Basquette, O., Bera, A., Chan, J. Y. H., Chingangbam, P., Cruz, H. A. G., Dasgupta, S., Datta, K. K., Fialkov, A., **Giri, S. K.**, Han, Q., Iliev, I. T., Li, B., Minoda, T., Mittal, S., Muñoz, J. B., Naik, S. S., Raste, J., Schneider, A., Sikder, S., Wu, K., Xu, Y., Yue, B., Zhang, M., Zhao, M.-L., 2026. High-Redshift Signatures from the Cosmic Dawn and the Epoch of Reionization. *Advancing Astrophysics with the SKA – II (AASKAII)*, Report number [AASKAII/Barkana01](#)
7. Bull, P., Burba, J., Bull, P., Ceccotti, E., Chakraborty, A., Chapman, E., Choudhuri, S., Datta, A., Elahi, K. M. A., **Giri, S. K.**, Jelić, V., Mao, Y., Mertens, F., Munshi, S., Nunhokee, R., Offringa, A. R., Pal, S. K., Sagar, R., Sims, P. H., Shan, H., Ito, T., Tripathi, A., Tolley, E., Zhang, L., Zhu, Z., 2026. Foreground characterization and mitigation in the observations of the CD/EoR with the SKA. *Advancing Astrophysics with the SKA – II (AASKAII)*, Report number [AASKAII/Burba01](#)
8. Chakraborty, A., Choudhury, T. R., Datta, K. K., Dayal, P., Dhandha, J., Gagnon-Hartman, S., **Giri, S. K.**, Gorce, A., Heneka, C., Hutter, A., Maity, B., Majumdar, S., Mesinger, A., Moriwaki, K., Murmu, C. S., Qin, Y., Yoshiura, S., Zackrisson, E., 2026. Synergies for the Epoch of Reionization and Cosmic Dawn. *Advancing Astrophysics with the SKA – II (AASKAII)*, Report number [AASKAII/Chakraborty01](#)
9. **Giri, S. K.**, and Mertens, F. G., 2026. Impact of 21-cm foreground mitigation strategies on reionization power spectrum constraints. [MNRAS](#), **550**(2), [stag1258](#).
10. Torkamani, M., Reischke, R., Kovač, M., Nicola, A., Bucko, J., Refregier, A., **Giri, S. K.**, Schneider, A., Hagstotz, S., 2026. Baryonification III: An accurate analytical model for the dispersion measure probability density function of fast radio bursts. *OJAp*, **9**, [arXiv:2601.18784](#)
11. Cerardi, N., **Giri, S. K.**, Bianco, M., Piras, D., et al., 2026. Implicit inference of the reionization history with higher-order statistics of the 21-cm signal. [MNRAS](#), **549**(1), [stag779](#).
12. Schwandt, T. P., Georgiev, I., **Giri, S. K.**, Mellema, G. and Iliev, I. T., 2025. Impact of anisotropic photon emission from sources during the epoch of reionisation. [MNRAS](#), **547**(2), [stag129](#).

13. Kovač, M., Nicola, A., Bucko, J., Schneider, A., Reischke, R., **Giri, S. K.**, Teyssier, R., Schaller, M., Schaye, J., 2025. Baryonification II: Constraining feedback with X-ray and kinematic Sunyaev-Zel'dovich observations. *JCAP*, 2025(11), 43.
14. Schneider, A., Kovač, M., Bucko, J., Nicola, A., Reischke, R., **Giri, S. K.**, Teyssier, R., Tröster, T., Refregier, A., Schaller, M., Schaye, J., 2025. Baryonification: An alternative to hydrodynamical simulations for cosmological studies. *JCAP*, 2025(11), 46.
15. **Giri, S. K.**, Kakichii, K., Bianco, M. and Meerburg, P. D., 2025. Mapping neutral islands during end stages of reionization with photometric intergalactic medium tomography. *MNRAS*, 544(4), 3076-3093.
16. Ceccotti, E., Offringa, A. R., Mertens, F. G., Koopmans, L. V. E., Munshi, S., Chege, J. K., Acharya, A., Brackenhoff, S. A., Chapman, E., Ciardi, B., Ghara, R., Ghosh, S., **Giri, S. K.**, Höfer, C., Hothi, I., Mellema, G., Mevius, M., Pandey, V. N., Zaroubi, S., 2025. First upper limits on the 21-cm signal power spectrum of neutral hydrogen at $z=9.16$ from the LOFAR 3C196 field. *MNRAS*, 544(1), 1255-1283.
17. Bonaldi, A., Hartley, P., Braun, R., Purser, S., Acharya, A., Ahn, K., Aparicio Resco, M., Bait, O., Bianco, M., Chakraborty, A., Chapman, E., Chatterjee, S., Chege, K., Chen, H., Chen, X., Chen, Z., Conaboy, L., Cruz, M., Darriba, L., De Santis, M., Denzel, P., Diao, K., Feron, J., Finlay, C., Gehlot, G., Ghosh, S., **Giri, S. K.**, ..., 2025. Square Kilometre Array Science Data Challenge 3a: foreground removal for an EoR experiment. *MNRAS*, 543(2), 1092-1119.
18. **Giri, S. K.**, 2025. AstronomyCalc: A python toolkit for teaching Astronomical Calculations and Data Analysis methods. *JOSE*, 8(87), 261.
19. Ghara, R., Zaroubi, S., Ciardi, B., Mellema, G., **Giri, S. K.**, Mertens, F. G., Mevius, M., Koopmans, L. V. E., Iliev, I. T., Acharya, A., Brackenhoff, S. A., Ceccotti, E., Georgiev, I., Ghosh, S., Hothi, I., Höfer, C., Ma, Q., Munshi, S., Offringa, A. R., Shaw, A. K., Yatawatta, S., 2025. Constraints on the state of the IGM at $z \sim 8 - 10$ using redshifted 21-cm observations with LOFAR. *A&A*, 699, A109.
20. Mertens, F. G., Mevius, M., Koopmans, L. V. E., Offringa, A. R., Zaroubi, S., Acharya, A., Brackenhoff, S. A., Ceccotti, E., Chapman, E., Chege, K., Ciardi, B., Ghara, R., Ghosh, S., **Giri, S. K.**, Hothi, I., Höfer, C., Iliev, I. T., Jelić, V., Ma, Q., Mellema, G., Munshi, S., Pandey, V. N., Yatawatta, S., 2025. Deeper multi-redshift upper limits on the Epoch of Reionization 21-cm signal power spectrum from LOFAR between $z = 8.3$ and $z = 10.1$. *A&A*, 698, A186.
21. Georgiev, I., Mellema, G., and **Giri, S. K.**, 2025. The forest at EndEoR: The effect of Lyman Limit Systems on the End of Reionisation. *MNRAS*, 536(4), 3689-3706.
22. Nebrin, O., Smith, A., Lorinc, K., Hörnquist, J., Larson, Å., Mellema, G., **Giri, S. K.**, 2025. Lyman- α feedback prevails at Cosmic Dawn: Implications for the first galaxies, stars, and star clusters. *MNRAS*, 537(2), 1646-1687.
23. Gao, L.-Y., Koopmans, L. V. E., Mertens, F. G., Munshi, S., Li, Y., Brackenhoff, S. A., Ceccotti, E., Chege, J. K., Acharya, A., Ghara, R., **Giri, S. K.**, Iliev, I. T., Mellema, G., Zhang, X., 2025. Extracting the Epoch of Reionization Signal with 3D U-Net Neural Networks Using a Data-driven Systematic Effect Model. *ApJ*, 988(1), 84.
24. Acharya, A., Ma, Q., **Giri, S. K.**, Ciardi, B., Ghara, R., Mellema, G., Zaroubi, S., Hothi, I., Iliev, I. T., Koopmans, L. V. E., Bianco, M., 2025. Exploring the effect of different cosmologies on the Epoch of Reionization 21-cm signal with POLAR. *MNRAS*, 543(2), 1058-1078.
25. Bianco, M., **Giri, S. K.**, Sharma, R., Chen, T., Krishna, S. P., Finlay, C., Nistane, V., Denzel, P., Santis, M. D., Ghorbel, H., 2025. Deep learning approach for identification of HII regions during reionization in 21-cm observations – III. image recovery. *MNRAS*, 541(1), 234-250.
26. Choudhury, M., Ghara, R., Zaroubi, S., Ciardi, B., Koopmans, L. V. E., Mellema, G., Shaw, A. K., Acharya, A., Iliev, I. T., Ma, Q.-B., **Giri, S. K.**, 2025. Inferring IGM parameters from the redshifted 21-cm Power Spectrum using Artificial Neural Networks. *JCAP*, 2025(06), 32.
27. Euclid Collaboration, Lesgourgues, J., Schwagereit, J., Bucko, J., Parimbelli, G., **Giri, S. K.**, ..., 2025. Euclid preparation. LVI. Sensitivity to non-standard particle dark matter model. *A&A*, 693, A249.

28. Euclid Collaboration,...**Giri, S. K.**, ..., 2025. Euclid. I. Overview of the Euclid mission. [A&A, 697, A1](#).
29. Schaeffer, T., **Giri, S. K.**, and Schneider, A., 2024. Testing common approximations to predict the 21cm signal at the Epoch of Reionization and Cosmic Dawn. [PRD, 110, 023543](#).
30. **Giri, S. K.**, Bianco, M., Schaeffer, T., Iliev, I. T., Mellema, G., and Schneider, A., 2024. The 21-cm signal during the end stages of reionization. [MNRAS, 533\(2\), 2364-2378](#).
31. Dayal, P. and **Giri, S. K.**, 2024. Warm dark matter constraints from the JWST. [MNRAS, 528\(2\), 2784-2789](#).
32. Ghara, R., Shaw, A. K., Zaroubi, S., Ciardi, B., Mellema, G., Koopmans, L. V. E., Acharya, A., Choudhury, M., **Giri, S. K.**, Iliev, I. T., Ma, Q. and Mertens, F. G., 2024. Probing the intergalactic medium during the Epoch of Reionization using 21-cm signal power spectra. [A&A, 687, A252](#).
33. He, Y., **Giri, S. K.**, Sharma, R., Mtchedlidze, S., and Georgiev, I., 2024. Inverse Gertsenshtein effect as a probe of high-frequency gravitational waves. [JCAP, 2024\(05\), 051](#).
34. Hirling, P., Bianco, M., **Giri, S. K.**, Iliev, I. T., Mellema, G., and Kneib, J. P., 2024. pyC²Ray: A flexible and GPU-accelerated Radiative Transfer Framework for Simulating the Cosmic Epoch of Reionization. [A&C, 48, 100861](#).
35. Peters, F. H., Schneider, A., Bucko, J., **Giri, S. K.**, and Parimbelli, G., 2024. Constraining Hot Dark Matter Sub-Species with Weak Lensing and the Cosmic Microwave Background Radiation. [A&A, 687, A161](#).
36. Bucko, J., **Giri, S. K.**, Peters, F. H. & Schneider, A., 2024. Probing the two-body decaying dark matter scenario with weak lensing and the cosmic microwave background. [A&A, 683, A152](#).
37. Schaeffer, T., **Giri, S. K.** and Schneider, A., 2023. BEoRN: a fast and flexible framework to simulate the epoch of reionization and the cosmic dawn. [MNRAS, 526\(2\), 2942-2959](#).
38. Acharya, A., Mertens, F., Ciardi, B., Ghara, R., Koopmans, L.V.E., **Giri, S. K.**, Hothi, I., Ma, Q.-B., Mellema, G. and Munshi, S., 2023. 21-cm Signal from the Epoch of Reionization: A Machine Learning upgrade to Foreground Removal with Gaussian Process Regression. [MNRAS, 527\(3\), 7835-7846](#).
39. Bianco, M., **Giri, S. K.**, Prelogović, D., Chen, T., Mertens, F. G., Tolley, E., Mesinger, A., and Kneib, J.-P., 2024. Deep learning approach for identification of HII regions during reionization in 21-cm observations – II. foreground contamination. [MNRAS, 528\(3\), 5212-5230](#).
40. Nebrin, O., **Giri, S. K.** and Mellema, G., 2023. Starbursts in low-mass haloes at Cosmic Dawn. I. The critical halo mass for star formation. [MNRAS, 524\(2\), 2290-2311](#).
41. Schneider, A., Schaeffer, T. and **Giri, S. K.**, 2023. Cosmological forecast of the 21-cm power spectrum using the halo model of reionization. [PRD, 108, 043030](#).
42. Bucko, J., **Giri, S. K.** and Schneider, A., 2023. Constraining dark matter decays with cosmic microwave background and weak lensing shear observations. [A&A, 672, A157](#).
43. **Giri, S. K.**, Schneider, A., Maion, F. and Angulo, R. E., 2023. Suppressing variance in 21-cm signal simulations during reionization. [A&A, 669, A6](#).
44. Gan, H., Mertens, F. G., Koopmans, L. V. E., Offringa, A. R., Mevius, M., Pandey, V. N., Brackenhoff, S. A., Ceccotti, E., Ciardi, B., Gehlot, B. K., Ghara, R., **Giri, S. K.**, Iliev, I. T., and Munshi, S., 2023. Assessing the impact of two independent direction-dependent calibration algorithms on the LOFAR 21-cm signal power spectrum. [A&A, 669, A20](#).
45. Gan, H., Koopmans, L. V. E., Mertens, F. G., Mevius, M., Offringa, A. R., Ciardi, B., Gehlot, B. K., Ghara, R., Ghosh, A., **Giri, S. K.**, Iliev, I. T., Mellema, G., Pandey, V. N., Zaroubi, S., 2022. Statistical analysis of the causes of excess variance in the 21 cm signal power spectra obtained with the Low-Frequency Array. [A&A, 663, A9](#).
46. Gehlot, B. K., Koopmans, L. V. E., Offringa, A. R., Gan, H., Ghara, R., **Giri, S. K.**, Kuiack, M., Mertens, F. G., Mevius, M., Mondal, R., Pandey, V. N., Shulevski, A., Wijers, R. A. M. J. and Yatawatta, S., 2022. Degree-Scale Galactic Radio Emission at 122 MHz around the North Celestial Pole with LOFAR-AARTFAAC. [A&A, 662, A97](#).

47. Mevius, M., Mertens, F., Koopmans, L. V. E., Offringa, A. R., Yatawatta, S., Brentjens, M. A., Chapman, E., Ciardi, B., Gan, H., Gehlot, B. K., Ghara, R., Ghosh, A., **Giri, S. K.**, Iliev, I. T., Mellema, G., Pandey, V. N., Zaroubi, S., 2022. A numerical study of 21-cm signal suppression and noise increase in direction-dependent calibration. *MNRAS*, 509(3), 3693-3702.
48. **Giri, S. K.** and Schneider, A., 2022. Imprints of fermionic and bosonic mixed dark matter on the 21-cm signal at cosmic dawn. *PRD*, 105, 083011.
49. Georgiev, I., Mellema, G., **Giri, S. K.** and Mondal, R., 2022. The large-scale 21-cm power spectrum from reionization. *MNRAS*, 513(4), 5109-5124.
50. Schneider, A., **Giri, S. K.**, Stefania, A. and Refregier, A., 2022. Constraining baryonic feedback and cosmology with weak-lensing, X-ray, and kinematic Sunyaev-Zeldovich observations. *MNRAS*, 514(3), 3802-3814.
51. Hothi, I., Chapman, E., Pritchard, J. R., Mertens, F. G., Koopmans, L. V. E., Ciardi, B., Gehlot, B. K., Ghara, R., Ghosh, A., **Giri, S. K.**, Iliev, I. T., Jelić, V., Zaroubi, S., 2021. Comparing Foreground Removal Techniques for Recovery of the LOFAR-EoR 21cm Power Spectrum. *MNRAS*, 500(2), 2264-2277.
52. Hubert, J., Schneider, A., Potter, D., Stadel, J., and **Giri, S.K.**, 2021. Decaying dark matter: simulations and weak-lensing forecast. *JCAP*, 2021(10), 040.
53. **Giri, S. K.** and Schneider, A., 2021. Emulation of baryonic effects on the matter power spectrum and constraints from galaxy cluster data. *JCAP*, 2021(03), 046.
54. Parimbelli, G., Scelfo, G., **Giri, S. K.**, Schneider, A., Archidiacono, M., Camera, S. and Viel, M., 2021. Mixed dark matter: matter power spectrum and halo mass function. *JCAP*, 2021(12), 044.
55. Ghara, R., **Giri, S. K.**, Ciardi, B., Mellema, G. and Zaroubi, S., 2021. Constraining the state of the intergalactic medium during the Epoch of Reionization using MWA 21-cm signal observations. *MNRAS*, 503(3), 4551-4562.
56. Bianco, M., Iliev, I.T., Ahn, K., **Giri, S. K.**, Mao, Y., Park, H. and Shapiro, P.R., 2021. The impact of inhomogeneous subgrid clumping on cosmic reionization II: modelling stochasticity. *MNRAS*, 504(2), 2443-2460.
57. Bianco, M., **Giri, S. K.**, Iliev, I.T. and Mellema, G., 2021. Deep learning approach for identification of HII regions during reionization in 21-cm observations. *MNRAS*, 505(3), 3982-3997.
58. **Giri, S. K.** and Mellema, G., 2021. Measuring the topology of reionization with Betti numbers. *MNRAS*, 505(2), 1863-1877.
59. Schneider, A., **Giri, S. K.** and Mirocha, J., 2021. A halo model approach for the 21-cm power spectrum at cosmic dawn. *PRD*, 103(8), 083025.
60. Ross, H. E., **Giri, S. K.**, Mellema, G., Dixon, K. L., Ghara, R. and Iliev, I. T., 2021. Redshift-space distortions in simulations of the 21-cm signal from the cosmic dawn. *MNRAS*, 506(3), 3717-3733.
61. Greig, B., Mesinger, A., Koopmans, L. V. E., Ciardi, B., Mellema, G., Zaroubi, S., **Giri, S. K.**, Ghara, R., Ghosh, A., Iliev, I. T., Mertens, F. G., Mondal, R., Offringa, A. R. and Pandey, V. N., 2021. Interpreting LOFAR 21-cm signal upper limits at $z \sim 9.1$ in the context of high- z galaxy and reionisation observations. *MNRAS*, 501(1), 1-13.
62. Mondal, R., Fialkov, A., Fling, C., Iliev, I. T., Barkana, R., Ciardi, B., Mellema, G., Zaroubi, S., Koopmans, L. V. E., Mertens, F. G., Gehlot, B. K., Ghara, R., Ghosh, A., **Giri, S. K.**, Offringa, A., Pandey, V. N., 2020. Tight Constraints on the Excess Radio Background at $z = 9.1$ from LOFAR. *MNRAS*, 498(3), 4178-4191.
63. Zackrisson, E., Majumdar, S., Mondal, R., Binggeli, C., Sahlén, M., Choudhury, T. R., Ciardi, B., Datta, A., Datta, K. K., Dayal, P., Ferrara, A., **Giri, S. K.**, Maio, U., Malhotra, S., Mellema, G., Mesinger, A., Rhoads, J., Rydberg, C.-E., Shimizu, I., 2020. Bubble mapping with the Square Kilometer Array-I. Detecting galaxies with Euclid, JWST, WFIRST and ELT within ionized bubbles in the intergalactic medium at $z > 6$. *MNRAS*, 493(1), 855-870.

64. **Giri, S. K.**, Mellema, G. and Jensen, H., 2020. Tools21cm: A python package to analyse the large-scale 21-cm signal from the Epoch of Reionization and Cosmic Dawn. [JOSS, 5\(52\), 2363](#).
65. Ghara, R., **Giri, S. K.**, Mellema, G., Ciardi, B., Zaroubi, S., Iliev, I. T., Koopmans, L. V. E., Chapman, E., Gazagnes, S., Gehlot, B. K., Ghosh, A., Jelić, V., Mertens, F. G., Mondal, R., Schaye, J., Silva, M. B., Asad, K. M. B., Kooistra, R., Mevius, M., Offringa, A. R., Pandey, V. N., Yatawatta, S., 2020. Constraining the intergalactic medium at $z \approx 9.1$ using the LOFAR epoch of reionization observation. [MNRAS, 493\(4\), 4728-4747](#).
66. Mertens, F. G., Mevius, M., Koopmans, L. V. E., Offringa, A. R., Mellema, G., Zaroubi, S., Brentjens, M. A., Gan, H., Gehlot, B. K., Pandey, V. N., Sardarabadi, A. M., Vedantham, H. K., Yatawatta, S., Asad, K. M. B., Ciardi, B., Chapman, E., Gazagnes, S., Ghara, R., Ghosh, A., **Giri, S. K.**, Iliev, I. T., Jelić, V., Kooistra, R., Mondal, R., Schaye, J., Silva, M. B., 2020. Improved upper limits on the 21-cm signal power spectrum of neutral hydrogen at $z \approx 9.1$ from LOFAR. [MNRAS, 493\(2\), 1662-1685](#).
67. **Giri, S. K.**, Zackrisson, E., Binggeli, C., Pelckmans, K. and Cubo, R., 2020. Identifying reionization-epoch galaxies with extreme levels of Lyman continuum leakage in James Webb Space Telescope surveys. [MNRAS, 491\(4\), 5277-5286](#).
68. **Giri, S. K.**, D'Aloisio, A., Mellema, G., Komatsu, E., Ghara, R. and Majumdar, S., 2019. Position-dependent power spectra of the 21-cm signal from the epoch of reionization. [JCAP, 2019\(02\), 058](#).
69. **Giri, S. K.**, Mellema, G., Aldheimer, T., Dixon, K. L. and Iliev, I. T., 2019. Neutral island statistics during reionization from 21-cm tomography. [MNRAS, 489\(2\), 1590-1605](#).
70. Watkinson, C. A., **Giri, S. K.**, Ross, H. E., Dixon, K. L., Iliev, I. T., Mellema, G. and Pritchard, J. R., 2019. The 21cm bispectrum as a probe of non-Gaussianities due to X-ray heating. [MNRAS, 482\(2\), 2653-2669](#).
71. **Giri, S. K.**, Mellema, G. and Ghara, R., 2018. Optimal identification of HII regions during reionization in 21-cm observations. [MNRAS, 479\(4\), 5596-5611](#).
72. Ghara, R., Mellema, G., **Giri, S. K.**, Choudhury, T. R., Datta, K. K. and Majumdar, S., 2018. Prediction of the 21-cm signal from reionization: comparison between 3D and 1D radiative transfer schemes. [MNRAS, 476\(2\), 1741-1755](#).
73. **Giri, S. K.**, Mellema, G., Dixon, K. L. and Iliev, I. T., 2018. Bubble size statistics during reionization from 21-cm tomography. [MNRAS, 473\(3\), 2949-2964](#).

Software/Code.....

74. Hirling, P., Bianco, M., **Giri, S. K.**, Iliev, I. T., Mellema, G., Kneib, J-P, 2023. pyC2Ray: Python interface to C2Ray with GPU acceleration. [ASCL, ascl:2312.025](#).
75. Mellema, G., Iliev, I. T., **Giri, S. K.**, Bianco, M., 2023. C2-Ray3Dm: 3D version of C2-Ray for multiple sources, hydrogen only. [ASCL, ascl:2312.023](#).
76. **Giri, S. K.** and Schneider, A., 2023. BCMemu: Model baryonic effects in cosmological simulations. [ASCL, ascl:2308.010](#).
77. Bianco, M., Iliev, I. T., Ahn, K., **Giri, S. K.**, Mao, Y., Park, H. and Shapiro, P. R., 2023. SubgridClumping: Clumping factor for large low-resolution N-body simulations. [ASCL, ascl:2306.050](#).

Conference Proceedings.....

78. de Salis, E., De Santis, M., Piras, D., **Giri, S. K.**, Bianco, M., Cerardi, N., Denzel, P., Selcuk-Simsek, M., Hess, K. M., Toribio, M. C., Kirsten, F., Ghorbel, H., 2026. Exploring the Early Universe with Deep Learning. Progress in Artificial Intelligence, EPIA 2025. Lecture Notes in Computer Science, vol 16121, Springer, Cham.
79. **Giri S. K.**, Nebrin O. and Mellema G., 2023. Imprints of early Universe galaxy formation on the 21-cm signal at cosmic dawn. [The URSI GASS 2023 Proceedings](#).
80. **Giri S. K.**, Zackrisson E., Binggeli C., Pelckmans K., Cubo R. and Mellema G., 2017. Constraining Lyman continuum escape using Machine Learning. [Proceedings of the International Astronomical Union, 12\(S333\), 254-258](#).

81. Mellema G., **Giri S.** and Ghara R., 2017. Analysis of 21-cm tomographic data. [Proceedings of the International Astronomical Union, 12\(S333\), 26-29.](#)
82. Ghara R., Choudhury T. R., Datta K. K., Mellema G., Choudhuri S., Majumdar S. and **Giri S. K.**, 2017. Prospects of detection of the first sources with SKA using matched filters. [Proceedings of the International Astronomical Union, 12\(S333\), 122-125.](#)

[Upcoming Publications.....](#)

83. **Giri, S. K.**, Meerburg, P. D., Koopmans, L. V. E., Fourier phase information in the 21-cm signal during reionization.